

Detailed Requirements

Functional Requirements

These define what Roomies must do to meet user needs.

1. User Account Management

The system shall allow users to create and authenticate an account using email and password.

2. User Profile Input

The system shall allow users to input:

- Personality traits
- Living habits
- Preferences (location, dates, room type, budget)

3. Roommate Matching

The system shall match users using an AI-based compatibility algorithm based on personal and lifestyle data.

4. Messaging

The system shall enable real-time messaging between matched users.

5. Task & Grocery Management

The system shall allow users to create, assign, and track errands and grocery requests in a shared list.

6. Payments

The system shall support peer-to-peer payment transactions for shared expenses (e.g., bills, groceries).

7. AI Chatbot Support

The system shall provide users with an AI chatbot for assistance, conflict resolution, and FAQs.

Non-Functional Requirements

These define how Roomies should perform.

1. Cross-Platform Compatibility

The system shall run smoothly on Android, iOS, and web browsers.

2. Data Security

The system shall ensure all user data and communications are protected with end-to-end encryption.

3. Performance

The system shall maintain a response time of under 2 seconds for all major interactions.

4. Availability

The system shall maintain 99.9% uptime, excluding scheduled maintenance.

5. Usability

The user interface shall be intuitive, requiring no more than 3 steps to access core functions.

6. Scalability

The system shall support up to 100,000 concurrent users without performance degradation.